AI & MACHINE LEARNING PROJECT

Movie Recommendation System using AI

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Analysis of Content-Based Filtering via NLP

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The Problem Statement

Choice Overload

With thousands of movies available, users struggle to find content they enjoy, leading to decision fatigue and reduced platform engagement.

The Goal

Build an automated engine that understands movie "DNA" (genres/plots) to provide highly personalized suggestions through Content-Based Filtering.

System Architecture

Data Input

Raw Movie Metadata (Titles, Genres, Overviews)

Processing Engine

Text Cleaning (Stop-words) & Vectorization

Output

Ranked Top 5–10 Similar Movies

The Math Behind the Magic

TF-IDF Vectorization

Calculates word importance by penalizing common words and rewarding unique plot markers.

```
Score = TF(t,d) * IDF(t)
```

Cosine Similarity

Measures the angle between two movie vectors to determine contextual proximity.

```
cos(θ) = (A·B) / (||A|| ||B||)
```

Implementation Stack

Python 3.x

Core Language for ML & Scripting

Scikit-Learn

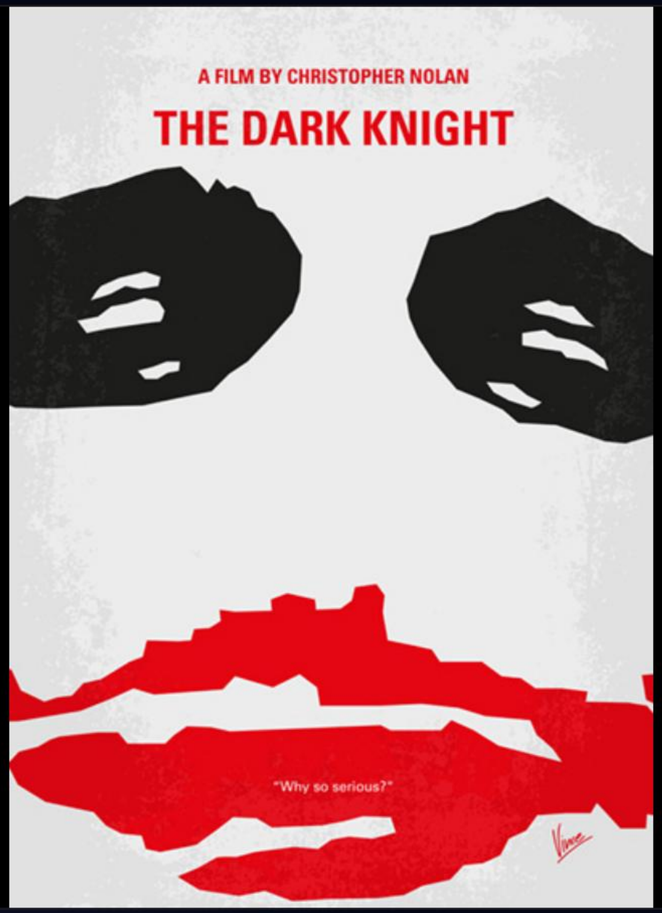
TfidfVectorizer & Linear Kernel

Pandas/NumPy

Efficient Matrix Operations

Environment: Jupyter Notebook / VS Code

Demonstration & Results



Input: The Dark Knight

Logic: Analyzing keywords: "Crime", "DC Comics", "Grim", "Batman".

Top Recommendations:

- 1. The Dark Knight Rises
- 2. Joker
- 3. The Godfather
- 4. Batman Begins

Result: System successfully identifies thematic "Grim/Crime Drama" proximity.

Advantages & Limitations

Advantages

- No "Cold Start" for items: Works immediately for new content.
- High Transparency: We know exactly why a movie was suggested.
- Privacy: Doesn't require user-to-user personal data.

Limitations

- "Filter Bubble": Limited to recommending similar items (lack of serendipity).
- Metadata Quality: Relies strictly on high-quality text descriptions.

Conclusion & Future Scope

Conclusion

We successfully built a functional NLP-based recommender using Python that leverages plot DNA to bridge the gap between content and user interest.

Future Scope

- **Hybrid Model:** Integrating Collaborative Filtering.
- **Web Interface:** UI using Streamlit or Flask.
- **Deployment:** Model serving via AWS or GCP.

Questions?

Thank you for your time.

Inquiry focus: TF-IDF Logic | Similarity Matrix | Vector Space

Image Sources

https://www.chungkong.nl/wp-content/uploads/2013/12/No245-My-Dark-Knight-minimal-movie-posterthumb.jpg

Source: www.chungkong.nl